Reachability in Two-Parametric Clock Systems

Internship Report

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Timed automata, introduced in the 1990s by Alur and Dill [AD90], extend finite automata with real-valued clocks. These clocks can be compared against constants. Timed automata provide a fundamental formalism for modeling and analyzing real-time systems. One of their main strengths is their robust algorithmic behavior. In particular, Alur and Dill showed that the reachability (or emptiness) problem is decidable and PSPACE-complete [AD94].

To model systems with partially unknown timing constraints, Alur, Henzinger, and Vardi introduced parametric timed automata (PTAs) [AHV93]. In PTAs, some constants appearing in guards are replaced by parameters. These parameters range over non-negative integers. They make it possible to represent under-specified systems, such as embedded systems interacting with uncertain environments. The reachability problem asks whether there exists a valuation of the parameters such that an accepting state is reachable in the resulting timed automaton.

A clock is called *parametric* if it is compared with at least one parameter. The decidability and complexity of the reachability problem for PTAs strongly depend on the number of parametric clocks. Alur, Henzinger, and Vardi proved that reachability is decidable for PTAs with a single parametric clock [AHV93]. Later work refined this result by establishing tight complexity bounds. The problem is PSPACE-complete with one clock and no parameter. It becomes EXPSPACE-complete as soon as additional clocks are allowed [BO17, GH20]. In contrast, the problem becomes undecidable with three parametric clocks, even if there is only one parameter [AHV93].

The case of two parametric clocks has remained a challenging open problem for many years. Bundala and Ouaknine [BO17] proved decidability when there is a single parameter. However, the general case with several parameters is still open. More recently, Göller and Hilaire [GH20] clarified the complexity of related settings. They proved EXPSPACE-completeness for systems with two clocks and one parameter. Despite these advances, the decidability of unrestricted PTAs with two parametric clocks remains unknown.

In this report, we study parametric systems with two clocks and two parameters. Our goal is to generalize this framework to an arbitrary number n of parameters. We will show that the main obstacles of the original proof remain in this more general setting.

To address this question, we consider an equivalent model: parametric one-counter automata (POCAs) with two parameters [BO17]. We investigate two possible generalizations. The first approach reduces a POCA with n parameters to a POCA with $n-1$ parameters. It is based on the proof of [GH20]. The second approach expresses the existence of a run in a POCA as a formula in Presburger arithmetic. It relies on the techniques developed in [Haa12, HKOW09].

Section 1 introduces the necessary notions and terminology. Section 2 presents the main result of the first approach. In Section 4, we reduce the problem to a simpler setting. In this setting, the POCA counter is bounded by a smaller value, and some transitions satisfy additional restrictions. We then reduce the problem to the satisfiability of a formula in the existential fragment of Presburger arithmetic with divisibility.

<table>
<tr><th>Params\ Clocks</th><th>1 clock</th><th>2 clocks</th><th>≥3 clocks</th></tr>
<tr><td>0</td><td colspan="3">PSPACE-complete [AD94]</td></tr>
<tr><td>1</td><td rowspan="2">PSPACE-complete [AHV93]</td><td>EXPSPACE-complete [BO17, GH20]</td><td rowspan="2">Undecidable [AHV93]</td></tr>
<tr><td>≥ 2</td><td>Open</td></tr>
</table>

Table 1: State of the art for the emptiness problem.

Contents

1 Preliminaries

This report uses notions such as nondeterminism and EXPSPACE complexity. If the reader is not familiar with these concepts, we refer to [Pap94, AB09] for further details.

Moreover, for a set E, we denote by $\max_{\leq} E$ its maximal element with respect to the order $\leq$. When the order is clear from the context, we simply write $\max E$. We also denote by $\mathrm{lcm}(n)$ the least common multiple of the set $[1, n]$. Remained that $\mathrm{lcm}(a, b) = \frac{a \cdot b}{\gcd(a,b)}$. Moreover, the ring of natural integers modulo c is denoted by $\mathbb{N}_c$. Furthermore, X denotes the set of clocks of a parametric timed automaton. We write $\mathcal{G}(X, P, \mathbb{Z})$ for the set of guards of the form $x \bowtie n$, where $x \in X$, $n \in \mathbb{N} \cup P$ and $\bowtie \in \{<, \leq, =, \geq, >\}$. We denote by 2^X the power set of X, i.e., the set of all subsets of clocks. Furthermore, we consider clocks with integer semantics, i.e., valuations over $\mathbb{N}$ rather than $\mathbb{R}$, since it is shown in [AD94] that the reachability problem is equivalent in both settings. An accepting run over PTA is a run starting in q_0 and ending in a final state. A valuation of parameters is a function ν from P to $\mathbb{N}$, while a clock valuation is a function δ from X to $\mathbb{N}$. For every guard $g = x \bowtie p$ with $x \in X$ and $p \in P \cup \mathbb{N}$, we write $\delta \vDash_\nu g$ if $\delta(x) \bowtie \nu(p)$ with $p \in P$ or if $\delta(x) \bowtie p$ with $p \in \mathbb{N}$. A configuration over a PTA is a pair (q, δ) where q is a state and δ is a clock valuation. Finally, we denote by Const the set of constants appearing on the transitions of the automaton.

1.1 Presburger Arithmetic

In this report, we use the existential fragment of Presburger arithmetic with divisibility, denoted by $\exists$PAD. It is a first-order logical theory over the structure $\langle \mathbb{N}, <, +, |, 0, 1 \rangle$, where $|$ denotes the standard divisibility relation on natural numbers. An existential PAD formula is of the form $\exists x_1, \ldots, x_k.\ \varphi$, where φ is quantifier-free. For instance, the congruence relation $x \equiv y \pmod z$ can be expressed in $\exists$PAD as $\exists p.\ (z \mid p \ \wedge\ x - p = y \ \wedge\ y < z)$. A set $S \subseteq \mathbb{N}^k$ is said to be $\exists$PAD-definable if there exists a finite set R of $\exists$PAD formulae with free variables $y_1, \ldots, y_k$ such that $(n_1, \ldots, n_k) \in S \iff \bigvee_{\varphi \in R} \varphi(n_1, \ldots, n_k)$. Moreover, $\exists$PAD-definable sets are closed under union, intersection, and projection.

1.2 Parametric timed automata

We start by defining the model of parametric timed automata (PTAs) and the associated emptiness problem.

Definition 1.1. *A parametric timed automata (PTA) is a tuple $\mathcal{A} = (Q, q_0, F, X, P, \rightarrow)$ where:*

- *Q is a finite set of states;*
- *$q_0 \in Q$ is the initial state;*
- *$F \subseteq Q$ is the set of final states;*
- *X is a finite set of clocks;*
- *P is a finite set of parameters;*
- *$\rightarrow \subseteq Q \times \mathcal{G}(X, P, \mathbb{Z}) \times 2^X \times Q$.*

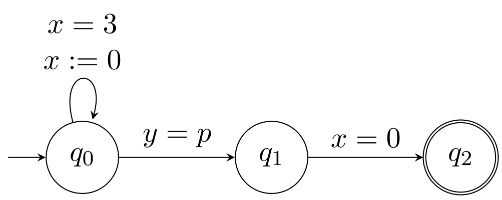


Figure 1: Example of a parametric timed automaton.

For example, the above figure 1 represents a PTA with two parametric clocks x, y and one parameter p. A parameter valuation ν witnesses that reachability holds for this PTA if, and only if, $\nu(p) = 3\mathbb{N}$.
In the following, we use the notation (n, k)-PTA to denote a PTA with n clocks and k parameters. A run is said to be accepting if it starts in the initial state, ends in an accepting state, and satisfies all the guards along the run. We use $\Delta(\pi)$ to denote the difference between the end and the beginning of the counter value along a run π.

The goal of this work is to study the emptiness problem for $(2, 2)$-PTAs and to generalize the results to the case of $(2, k)$-PTAs for all k.

Definition 1.2. *[AD94] The emptiness problem for (n, k)-PTAs is defined as follows: given a (n, k)-PTA $\mathcal{A}$, does there exist a valuation of the parameters such that $\mathcal{A}$ admits an accepting run?*

1.3 Parametric one counter automata

To address the problem of reachability in PTAs, we reduce it to the reachability problem in parametric one counter automata (POCAs) as defined in [BO17]. Hence, we first introduce the model of POCAs and the associated reachability problem.

Definition 1.3. *([BO17]) A parametric one counter automaton (POCA) is a tuple $\mathcal{C} = (Q, q_0, F, P, \rightarrow)$ where:*

- *Q is a finite set of states;*
- *$q_0 \in Q$ is the initial state;*
- *$F \subseteq Q$ is the set of accepting states;*
- *P is a finite set of parameters;*
- *$\rightarrow \subseteq Q \times (\mathcal{G}(P, \mathbb{Z}) \cup \mathcal{U}(P, \mathbb{N})) \times Q$.*

We will use $\mathcal{G}(P, \mathbb{Z})$ to denote all guards of the form $\bowtie n$ where $n \in \mathbb{Z} \cup P$ and $\mathcal{U}(P, \mathbb{N})$ to denote all updates of the form $\pm n$ where $n \in \mathbb{N} \cup P$. Moreover, a POCA can have a $+[0, p_i]$ transition, which is an update transition of the form $+n$ where $n \in [0, p_i]$ and a $\mod c$ transition, which is a guard transition which checks if the counter value is congruent to 0 modulo c where is a constante.

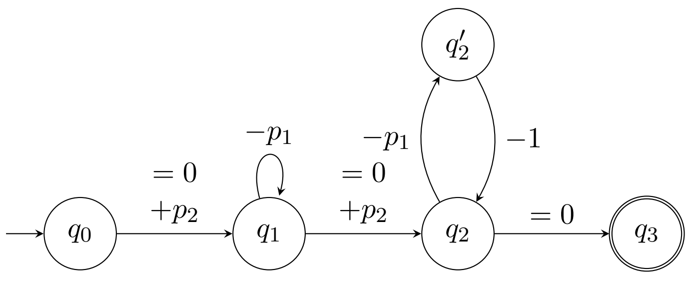


Figure 2: Example of a parametric one counter automaton.

For example, the above figure 2, represents a POCA with two parameters p_1 and p_2. We remark that we have an accepting run if p_2 is equal to 0 or if $p_1 | p_2$ and $p_1 + 1 | p_2$. Hence, a parameters valuation ν witnesses that reachability holds for this POCA if, and only if, $\nu(p_1, p_2) = \mathbb{N} \times \{0\} \cup \{(a, b) \in \mathbb{N}^2 | a | b \wedge a + 1 | b\}$.
The notation $counter(\pi)$ denote $\sum_{(\_,c) \in \pi} c$ and $\Delta(\pi)$ denote $y - x$ with x and y represents the first and the last counter value in π. We define the remaining vocabulary analogously to that used for PTAs. Moreover, $\vec{\pi}$ denotes the vector (a, b, c) such that $counter(\pi) = a \cdot p_2 + b \cdot p_1 + c$. A configuration of a POCA is a pair (q, c) where q is a state and c is the value of the counter.

Theorem 1.4. *([BO17]) For every $(2,k)$-PTA, one can effectively construct a POCA bounded by $2 \cdot p_{max}$ with an equivalent emptiness problem.*

The construction is detailed in [BO17]. A description of the gadgets responsible for modifying the counter value is provided in the appendix.

Theorem 1.5. *([BO17, HKOW09]) Given C a POCA and $s, t \in Q_C$, the set*

$$\Phi_0(C, s, t) = \{(x, y, \vec{n}) | (s, x) \to^* (t, y) \text{ in } C \text{ where } \vec{p} = \vec{n}\}$$

is $\exists$PAD definable.

Theorem 1.6. *([BO17]) Given C a POCA with one parameter and $s, t \in Q_C$, the set*

$$\Phi_1(C, s, t) = \{(x, y, n) | (s, x) \to^* (t, y) \text{ in } C \text{ where } p = n\}$$

is $\exists$PAD definable.

2 Reduction from n to $n-1$ parameters

In this section, we generalize the approach of [GH20], which establishes EXPTIME-completeness of the reachability problem for $(2,1)$-PTAs. Their reduction transforms reachability for one-parameter POCAs into the parameter-free setting. We extend this idea by reducing reachability for POCAs with n parameters to the case with $n-1$ parameters.

This generalization introduces two main challenges. First, in the multi-parameter setting, the reduction from PTAs to POCAs yields transitions of the form $+[0, p_i]$, allowing any value in the interval $[0, p_i]$ to be added to the counter. Second, the depumping argument becomes more intricate, as it must now account for interactions between multiple parameters.

2.1 Depumping lemma

The depumping lemma is the key ingredient of the reduction. Intuitively, if a run contains the same number of $+p_2$ and $-p_2$ transitions and induces a sufficiently large counter effect, then one can construct another run with the same structure but a slightly smaller effect.

In the statement below, N_i denotes the i-th parameter, and $\phi(\pi) \in \Lambda_8$ expresses that π is well-parenthesized with respect to $+p_2$ and $-p_2$ transitions. We also define

$$A = \{a \cdot p_1 + b \neq 0 | a, b \in [0, 17 \cdot n] \wedge a + b \leq 17 \cdot n\}$$

and

$$Z_C = \text{lcm}(\text{Const}_C) \qquad \Upsilon = (17 \cdot n)^2 \cdot \text{lcm}(A) \cdot 17 \cdot n \cdot p_1 \cdot Z_C \qquad \Gamma = \text{lcm}(A) \cdot Z_C.$$

Note that $\text{lcm}(A)$ is bounded by $(17 \cdot n \cdot (p_1 + 1))^{\frac{(17 \cdot n + 1) \cdot (17 \cdot n + 2)}{2}}$. Hence, $\text{lcm}(A)$ is polynomial in p_1.

Lemma 2.1 (Depumping lemma)**.** *For all $N_1, N_2 \in \mathbb{N}$, let π be an N_2-semirun such that $\phi(\pi) \in \Lambda_8$ and $|\Delta(\pi)| > \Upsilon$. Then there exists an N_2-semirun π' such that:*

- *if $\Delta(\pi) > \Upsilon$, then $\Delta(\pi') = \Delta(\pi) - \Gamma$;*
- *if $\Delta(\pi) < -\Upsilon$, then $\Delta(\pi') = \Delta(\pi) + \Gamma$.*

Proof. We consider the case $\Delta(\pi) > \Upsilon$ (the other being symmetric). Since

$$\Delta(\pi) > \Upsilon = 17 \cdot n \cdot \text{lcm}(17 \cdot n \cdot p_1) \cdot 17 \cdot n \cdot p_1 \cdot Z_C,$$

by the pigeonhole principle there exist at least $17 \cdot n \cdot \text{lcm}(17 \cdot n \cdot p_1)$ pairwise disjoint subsemiruns $\pi[a_i, b_i]$ such that $\Delta(\pi[a_i, b_i]) > 17 \cdot n \cdot p_1 \cdot Z_C$.

For each such segment, again by pigeonhole, there exist indices s_i, t_i with $a_i \leq s_i < t_i \leq b_i$, $q_{s_i} = q_{t_i}$, and $\pi[s_i, t_i]$ well-parenthesized. Moreover, $\Delta(\pi[s_i, t_i]) > (a \cdot p_1 + b) \cdot Z_C$ for some $a, b \in [1, 17 \cdot n]$ with $a + b \leq 17 \cdot n$.

Among the $17 \cdot n \cdot \mathrm{lcm}(17 \cdot n \cdot p_1)$ segments, there exist fixed a, b such that at least

$$\frac{\mathrm{lcm}(17 \cdot n \cdot p_1)}{a \cdot p_1 + b}$$

of them satisfy these same parameters. Let $\pi[a_{i_1}, b_{i_1}], \ldots, \pi[a_{i_k}, b_{i_k}]$ be these segments.

We then define π' by removing the corresponding subruns: $\pi' = \pi - [s_{i_1}, t_{i_1}] - \cdots - [s_{i_k}, t_{i_k}]$. This yields

$$\Delta(\pi') = \Delta(\pi) - (a \cdot p_1 + b) \cdot \frac{\mathrm{lcm}(A)}{a \cdot p_1 + b} \cdot Z_C = \Delta(\pi) - \Gamma,$$

as required. □

The lemma shows that any sufficiently large parameter value can be decreased in controlled steps of size Γ, until it lies in the bounded interval $[30(\Gamma + \Upsilon + 1) - \Gamma,\ 30(\Gamma + \Upsilon + 1)]$. Consequently, the parameter p_2 can be expressed as a polynomial in p_1 which completes the reduction step. Note that we cannot obtain a better asymptotic bound for p_2 as a function of p_1. Indeed, there exist POCAs with two parameters such that $p_2 > p_1^2 + p_1$. Consider the example in Figure 2, where $p_1 \mid p_2$ and $(p_1 + 1) \mid p_2$. Since $\gcd(p_1, p_1 + 1) = 1$, it follows that $p_1(p_1 + 1) \mid p_2$.

However, the decidability of the reachability problem for POCA with two parameters remains an open question when the counter is linearly bounded by one parameter and polynomially bounded by the other.

3 Transformation

Before starting, we first simplify both the counter bounds and the transition structure, and restrict the form of runs. Specifically, it is sufficient to consider runs whose counter values remain within $[0, p_1 + p_2]$. Moreover, the transition structure can be normalized so that $\pm p_2$ transitions occur only within the regions $[0, p_1]$ and $[p_2, p_1 + p_2]$, and all $+[0, i]$ and $\bmod c$ transitions are removed.

3.1 $\bmod c$ and $+[0, p_i]$ transitions

We show that there exists an equivalent POCA without $\bmod c$ transitions. The construction below generalizes the one from [BO17].

The main idea is to encode in the states the residue of each parameter and the counter modulo every constant in C. This allows us to construct an equivalent POCA C_{mod} as follows:

- $Q_{mod} = Q_C \times \mathbb{Z}_{c_1} \times \cdots \times \mathbb{Z}_{c_r}$ where $c_1, \ldots, c_r$ the constants of C and k the number of parameters;
- if $(q, \pm c, q') \in \rightarrow_C$, then $((q, v_1, \ldots, v_r), \pm c, (q', v_1 \pm c, \ldots, v_r \pm c)) \in \rightarrow_{C_{mod}}$;
- if $(q, \pm p_i, q') \in \rightarrow_C$, then $((q, v_1, \ldots, v_r), \pm p_i, (q', v_1 \pm d_{1_i}, \ldots, v_r \pm d_{r_i})) \in \rightarrow_{C_{mod}}$ where $d_{j_i} \equiv p_i \bmod c_j$;
- if $(q, \bmod c_i, q') \in \rightarrow_C$ and $v_i = 0$, then $((q, v_1, \ldots, v_r), +0, (q', v_1, \ldots, v_r)) \in \rightarrow_{C_{mod}}$;
- if $(q, \mathcal{G}(P, \mathbb{Z}), q') \in \rightarrow_C$, then $((q, v_1, \ldots, v_r), \mathcal{G}(P, \mathbb{Z}), (q', v_1, \ldots, v_r)) \in \rightarrow_{C_{mod}}$;

faire un schéma de la construction? ou voir seulement la partie choix de d_{i_j}.
And a new first transition to make a choice of each value d_{i_j} and test if $d_{i_j} \equiv p_i \bmod c_j$ to check that, we have to use the following gadget.

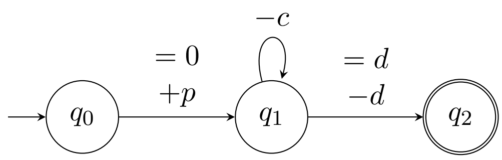


Figure 3: Gadget to check if $a \equiv d \bmod c$

Remark that there exists a run from (s, x) to (t, y) in C if, and only if, there exists a run from $((s, x \mod c_1, \ldots, x \mod c_r), x)$ to $((t, y \mod c_1, \ldots, y \mod c_r), y)$ in C_{mod}.

Thus, we can proof the following theorem.

Theorem 3.1. *Let C be a POCA with* mod$c$ *transitions Then*

$$\Phi_{\bmod c}(C, s, t) = \{(x, y, \vec{q}) | \exists \pi : (s, x) \to (t, y) \in C \textit{ where } \vec{p} = \vec{q}\}$$

is $\exists PAD$ definable.

Now, let us show that reachability for a POCA with $+[0, p_i]$ transitions is ∃PAD definable if, and only if, reachability problem for a POCA without $+[0, p_i]$ transitions is ∃PAD definable. This idea is the same that in [BO17]. We will use R to denote $lcm(Const_C)$.

Lemma 3.2. *Let C be a parametric bounded one-counter machine. If there is an accepting bounded run in C, then there is an accepting bounded run π such that at most $2 \cdot n \cdot R$ counter increments by $+[0, p_i]$ transitions are not in the set $\{+0, +1, \ldots, +R, +p_1 - R, \ldots, +p_1, \ldots, +p_r - R, \ldots, +p_r\}$. That is,*

$$|\{\, i \mid \exists j \leq r \ , \pi(i) = +[0, p_j] \textit{ and } R < \text{counter}(\pi(i + 1)) - \text{counter}(\pi(i)) < p_j - R\}| \leq 2 \cdot n \cdot R \cdot r.$$

Thus, with one parameter we get the following theorem.

Theorem 3.3. *([BO17]) Given a POCA C with $\equiv 0 \pmod c$ and $+[0, p]$ transitions and a single parameter p, let $s, t \in C$ be states. Then the set $\Phi_{+[0,p_i]}(C, s, t) = \{(x, y, q) \in \mathbb{N}^3 \mid \exists \pi : (s, x) \to^* (t, y) \text{ in } C,\ p = q\}$ is ∃PAD-definable.*

Hence, it is possible to repeat this process for every $+[0, p_i]$ transition and get the following result.

Theorem 3.4. *Given a POCA C with $\equiv 0 \pmod c$ and $+[0, p_i]$ transitions, let $s, t \in C$ be states. Then the set $\Phi_{+[0,P]}(C, s, t) = \{(x, y, \vec{q}) \mid \exists \pi : (s, x) \to^* (t, y) \text{ in } C,\ \vec{p} = \vec{q}\}$ is ∃PAD-definable if, and only if, reachability for C without* mod$c$ *and $+[0, p_i]$ is ∃PAD-definable.*

3.2 A better bound

As described in Theorem 1.4, we consider a POCA C whose counter is bounded by $2 \cdot p_{\max}$. In this section, we construct an equivalent POCA whose counter is bounded by $p_k + p_{k-1}$.

For clarity, we restrict our presentation to the case of two parameters, although the construction naturally generalizes to an arbitrary number n of parameters.

To define this new automaton, we introduce two auxiliary POCAs, denoted by $C_{<p_2}$ and $C_{\geq p_2}$.

The POCA $C_{\bowtie p_2}$ is constructed from C as follows:

- If $(q, \pm v, q') \in \to_C$ with $v \in Const \cup \{p_1\}$, then $(q, \pm v, q_{\text{bis}})$, $(q_{\text{bis}}, \bowtie p_2, q') \in \to_{C_{\bowtie p_2}}$.
- If $(q, \bowtie p_2, q') \in \to_C$, then $(q, +0, q') \in \to_{C_{\bowtie p_2}}$.
- If $\bowtie = <$ and $(q, g(v), q') \in \to_C$, with $v \in Const \cup \{p_1\}$ and $g \in \{<, \leq, =, \geq, >\}$, then $(q, g(v), q') \in \to_{C_{\bowtie p_2}}$.

Using these constructions, we define a new POCA, denoted by C_b, obtained from C, whose counter is bounded by $p_1 + p_2$.

The POCA C_b is defined from C as illustrated in Figure 4:

- $\rightarrow_{C_{<p_2}} \cup \rightarrow_{C_{\geq p_2}} \subseteq \rightarrow_{C_b}$.
- If $(q, +p_2, q') \in \rightarrow_C$, then $(q_{<p_2}, +0, q'_{\geq p_2}) \in \rightarrow_{C_b}$.
- If $(q, -p_2, q') \in \rightarrow_C$, then $(q_{\geq p_2}, -0, q'_{<p_2}) \in \rightarrow_{C_b}$.
- If $(q, +v, q') \in \rightarrow_C$ with $v \in \mathit{Const} \cup \{p_1\}$, then $(q_{<p_2}, \geq p_2, q_{\text{bis}})$, $(q_{\text{bis}}, -p_2, q_{\geq p_2}) \in \rightarrow_{C_b}$.
- If $(q, -v, q') \in \rightarrow_C$ with $v \in \mathit{Const} \cup \{p_1\}$, then $(q_{\geq p_2}, \leq 0, q_{\text{bis}})$, $(q_{\text{bis}}, +p_2, q_{<p_2}) \in \rightarrow_{C_b}$.

The resulting POCA has a counter ranging over $[-p_1,\ p_1 + p_2]$. To shift this interval to a non-negative range, we add p_1 before the initial state and increase every guard in $C_{<p_2}$ by p_1.

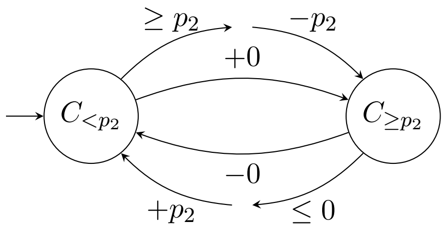


Figure 4: Description of the construction.

Remark that this new POCA can use $\pm p_2$ transitions only in $[0, p_1] \cup [p_2, p_2 + p_1]$. Now, let show that this new construction is equivalent to the previous one.

Lemma 3.5. *Given C a POCA and C_b is POCA as define above. These two following propositions are equivalent:*

- *There exists a run π from (s, x) to (t, y) in C.*
- *There exists a run π' from $(s, x \mod p_2)$ to $(t, y \mod p_2)$ in C_b.*

Now, in the remainder of this report, we assume that the POCA under study is bounded by $p_2 + p_1$, and that the $\pm p_2$ transitions can only occur when the counter lies in $[0, p_1]$ or $[p_2, p_2 + p_1]$. Thus, it is natural to study runs in terms of the bridges that move between the intervals $[0, p_1]$ and $[p_2, p_2 + p_1]$, and vice versa.

4 Reduction to ∃PAD

For the following part, we will use *bridge run* to denote run starting from $[2 \cdot p_1, 3 \cdot p_1]$ to $[p_2 - 2 \cdot p_1, p2 - p_1]$ where the counter between the first and the last transition is in $(3 \cdot p_1, p_2 - 2 \cdot p_1)$ or inversely. Note that $\pm p_2$ transitions are bridges. An *hill run* is a run starting and ending in $[0, 3 \cdot p_1]$ where its counter stays below $p_2 - 2 \cdot p_1$ in $C_{<p_2}$. An *valley run* is a run starting and ending in $[p_2 - 2 \cdot p_1, p_2 + p_1]$ where its counter stays above $3 \cdot p_1$ in $C_{\geq p_2}$.

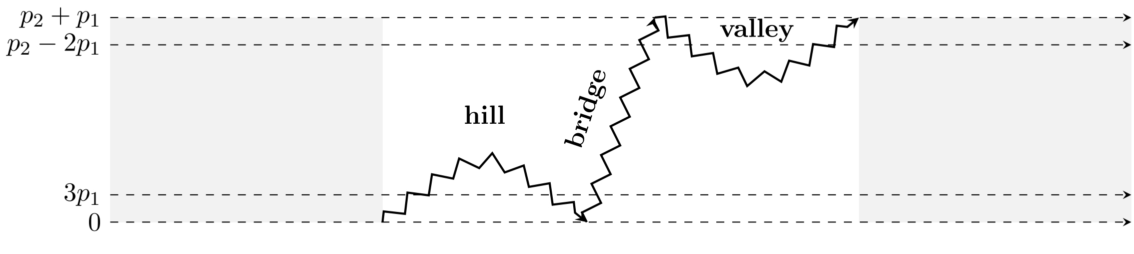


Figure 5: Schema of run vocabulary

We represent a run π from a configuration c_1 to a configuration c_2 as a concatenation of the form $\pi = \sigma_0\,\pi_1\,\sigma_1\,\pi_2 \cdots \pi_k\,\sigma_k$, where each σ_i is either a hill run or a valley run and π_i a bridge such that σ_i and $\pi_i \neq \epsilon$. Note that transitions of the form $\pm p_2$ occur only within the bridge segments π_i.

We say that a loop l is positive (resp. negative) if $\Delta(l) > 0$ (resp. $\Delta(l) < 0$). To do that, we have that each run in POCA can be represented as follows.

Lemma 4.1. *([Haa12, HKOW09]) Let C be a one-counter machine without upper bounds. For configurations c_1 and c_2, if there exists a run $\tau : c_1 \to c_2$, then there exists a run τ' from c_1 to c_2 such that $\tau' = \tau_1 \cdot \tau_2 \cdot \tau_3$ where:*

- *$min(counter(\tau')) \leq min(counter(\tau))$;*
- *There are no positive simple loops in τ_1 and no negative simple loops in τ_3;*
- *$\tau_2 = \alpha^a \to \sigma \to \beta^b$ where α and β are simple loops such that $\Delta(\beta) < 0 < \Delta(\alpha)$ and σ is of the form $(f + \sum k_i s_i)$ where f is a simple path, s_i are simple loops and $a, b, k_i \in \mathbb{N}$ and the set of edges $f \cup \{s_i\}$ is connected.*

Now, a type denote the following tuple $(\tau_1, \alpha, \sigma, \beta, \tau_3)$. Remark that the set of type is finite of size. We will use D to describe this size. We say that a run is *pumpable* if $\tau_2 \neq \epsilon$ and that a run is *linear* if for all ρ,σ respectively a positive and negative simple loop, $\rho \times \sigma = 0$ (that means the vectors representing their counter are collinear).

Hence, to aid the reader, we provide below a table 2 summarizing the main steps of the proof. The main idea is to get a better understanding of a run. To do that, we will study a run when the counter is linearly bounded by all parameters of the POCA. After that, we will see some properties of hill and valley subruns and bridge subruns. The goal is to identify finite behaviors that are definable in $\exists PAD$. Remark that we do not have found a way to study the case of linear bridge.

Options\ Subruns	**Hill and Valley**	**Bridge**
Non pumpable	bounded $(2 \cdot n \cdot p_1)$ (4.1)	does not exist (4.1)
Linear	bounded $(K \cdot p_1)$ ([BO17])	**open**
Non-linear	equivalence relation (4.7)	equivalence relation (4.5)
$\pm p_2$ **transitions**	does not exist (3.5)	reduce to $2 \cdot K \cdot p_1 + 1$ (4.9)

Table 2: Summarize of reduction proof.

This proof generalizes the approach of [BO17]and try to established decidability of the emptiness problem by reducing it to the satisfiability of a formula in ∃PAD.

Let us briefly review the proof in the single-parameter case. The main idea is to show that there exists an equivalent run consisting of two finite sets of subruns that are ∃PAD definable. In the first one, the counter is linearly bounded by the unique parameter and in the second one, the counter is not bounded. This property allows us to encode the acceptability of a run as an ∃PAD formula, since it is sufficient to existentially quantify the junctions between the bounded and unbounded subruns.

4.1 Linearly bounded by all parameters

In this section, we show that if $p_2 \leq K \cdot p_1$ for a constant K which is $O(p_1)$ described in Theorem 10.13 of [BO17], then the reachability problem for POCAs is ∃PAD definable. More precisely, if the POCA has n parameters, we will show the decidability proof when the counter is linearly bounded by every parameter by induction over n.

Thus, let C be a simple parametric bounded one-counter machine with parameters $p_1, \ldots, p_n$. For given bounds $k_1, \ldots, k_n \in \mathbb{N}$, we consider the following boundless condition: let π be an accepting run then $\forall i \in [1, n], \mathsf{counter}(\pi) \leq k_i p_i$. Let π be such a run. For every configuration along π we can uniquely decompose the counter value as $\mathsf{counter}(\pi) = q_1p_1 + \cdots + q_np_n + r$, where $0 \leq q_i < k_i$ for all $i \in [1, n]$ and $q_{i-1}p_{i-1} + \cdots + r < p_i$. Since each coefficient q_i is bounded, we construct a one-counter machine G that stores the tuple $(q_1, \ldots, q_n)$ in the control state and keeps the remainder r in the counter.

Lemma 4.2 ([LLT04]). *Let X be a one-counter machine. There exists a constant M (polynomial in $|X|$ and in the magnitude of the largest constant appearing in X) such that, for any configurations c_1 and c_2 of X, the following holds:*

- *Let $U = \min(\text{counter}(c_1), \text{counter}(c_2))$*
- *and $V = \max(\text{counter}(c_1), \text{counter}(c_2))$.*

Consequently, as long as $p_1 + M < \text{counter}(c_1), \text{counter}(c_2) < p_2 - M$, the runs $c_1 \to c_2$ in C_i correspond to runs in C.

We then apply the previous Lemma 4.2 to G and decompose π into segments that are either close to or far from a multiple of p_n More precisely, we write $\pi = \tau_0 \to \pi_1 \to \tau_1 \to \cdots \to \pi_\ell \to \tau_\ell$ such that:

- for every τ_i, we have $counter(\tau_i) \bmod p_n \in [0, M] \cup [p_n - M, p_n)$,
- and for every π_i, we have $counter(\pi_i) \bmod p_n \in (M, p_na - M)$.

Before proving the theorem 4.3, we have to explain how to decide the existence of the segments τ_i and π_i. Given C and bounds $k_1, \ldots, k_n$, we introduce a one-counter machine C_{mod}.

Let S be the set of states of C. The state space of C_{mod} is $S \times \{0, \ldots, k_n\}$. Intuitively, C_{mod} stores in its control state the bounded quotients $q_1, \ldots, q_n$, while the counter stores the remainder r. The transitions of C_{mod} are defined as follows:

- $((s, i), \pm c, (t, i))$ if $(s, \pm c, t)$ is a transition in C,
- $((s, i), \pm p, (t, i))$ if $(s, \pm p, t)$ is a transition in C and $p \neq p_n$,
- $((s, i), \pm 0, (t, i \pm 1))$ if $(s, \pm p_n, t)$ is a transition in C,
- $((s, i), \pm 0, (t, i))$ for $i \leq 1$ if $(s, \leq p_j, t)$ is a transition in C,
- $((s, i), \pm 0, (t, i))$ for $i > 0$ if $(s, \geq p_j, t)$ is a transition in C,

We can therefore state the following theorem; its proof is provided in the appendix.

Theorem 4.3. *Given C with n parameters, $k_i \in \mathbb{N}$, and states s, t of C, the set*

$$\Phi_2(C, s, t, k_1, \ldots, k_n) = \{(x, y) \mid \exists \pi : (s, x) \to (t, y) \text{ and } \Delta(\pi) \leq k_ip_i \text{ for all } i\}$$

is $\exists PAD$-definable.

Hence, for the following part, we can assume that $p_2 > K \cdot p_1$ with $K > 2 \cdot n$.

4.2 Bridges

The purpose of this section is to provide a better understanding of bridges within a run. For the next part, we will interest about bridges which are not a $\pm p_2$ transitions.

Lemma 4.4. *([BO17]) Let $\pi = \pi_1 \to \alpha^a \to (f + k_is_i) \to \beta^b \to \pi_3$ be a factored run. Let $u, v \in \{a, b, k_1, k_2, \ldots\}$ and let $\hat{u}, \hat{v} \in \mathbb{N}$. Then, if $u + \hat{u}, v + \hat{v} > 0$, there exists a run π', denoted by $\pi' = \pi(u \leftarrow u + \hat{u},\ v \leftarrow v + \hat{v})$, such that:*

- $\Delta(\pi') = \Delta(\pi) + \hat{u}\,\Delta(\sigma_u) + \hat{v}\,\Delta(\sigma_v)$,
- $\text{counter}(\pi') \geq \min\big(\text{counter}(\pi),\ \text{counter}(\pi) + \hat{u}\,\Delta(\sigma_u) + \hat{v}\,\Delta(\sigma_v)\big)$,

where σ_u and σ_v are the loops associated with u and v, respectively.

Now, let's consider a π_i run which is not a $\pm p_2$ transition. By 4.1, we can write $\pi_i = \pi_1\pi_2\pi_3$. By [BO17], if $|\Delta(\pi_i)| > n \cdot p_1$, then π_i is pumpable. Hence, by assumption $p_2 > n \cdot p_1$, then π_i is pumpable. We conclude that π_i is linear or non-linear run.

Now, the main idea is to group non-linear bridges in equivalence classes.

4.2.1 Non-linear bridges

Now, the following part show that we can group non-linear π_i in equivalence classes. As in [BO17], we have nonlinear run from configuration (s, x) to configuration (t, y). By definition, there is a positive loop ρ and a negative loop σ in the run such that $\rho \times \sigma \neq 0$. By modifying the number of times ρ and σ are traversed, we obtain runs from (s, x) to $(t, y \pm L)$ for some $L \in \mathbb{N}$ depending only on $\rho \times \sigma$. Repeating the process, we build nonlinear runs starting in (s, x) and finishing in configurations $(t, y + kL)$ for almost all valid $k \in \mathbb{Z}$. We thus obtain the following lemma.

Lemma 4.5. *Given a run $\pi = \sigma_0\pi_1\sigma_1\pi_2 \dots \pi_k\sigma_k$ where σ_i is a run starting and ending bellow $2 \cdot p_1$ (above $p_2 - p_1$) when the run stays bellow $p_2 - p_1$ (or above $2 \cdot p_1$) and π_i is a run starting bellow $2 \cdot p_1$ and ending above $p_2 - p_1$ (or starting above $2 \cdot p_1$ and ending bellow $p_2 - p_1$). Then, there exists an equivalent run $\pi' \sim \pi$ such that π' factors as:*

$$\sigma'_0\pi'_1\sigma'_1\pi'_2 \dots \pi'_m\sigma'_{g(m)}$$

where

- $\pi'_i : first(\pi_{f(i-1)}) \to last(\pi_{f(i)})$ *if π'_i is a nonlinear run;*
- $g : [1 \dots l] \to [1 \dots k]$ *is a strictly increasing function;*
- $m \leq 2 \cdot D \cdot n^5$.

Remark that a bridge contains only one parameter and can pass over every guard $\geq p_1$ and can not pass over guard $\leq p_2$. Thus, if we have a bridge from (s, x) to (t, y), we can call $Reach(G, s, t)(x, y)$ where G is a POCA with one parameter and without bound.

Theorem 4.6. *Given C and states $s, t \in Q_C$. The*

$$\Phi_{NL-B}(C, s, t) = \{(x, y, \vec{v}) | (s, x) \to^* (t, y) \text{ in } C \text{ without linear bridge,}$$

$$\text{where } \vec{v} = \gamma \text{ and } \gamma(p_2) > n \cdot \gamma(p_1)\}$$

is $\exists$PAD*-definable.*

The main difficulty arises at this point. We cannot group such runs into equivalence classes; therefore, the technique described previously cannot be applied in this setting.

4.2.2 Linear bridges

The main difficulty arises at this point. We cannot group such runs into equivalence classes; therefore, the technique described previously cannot be applied in this setting. Because, for a linear run, we have that for all $\rho \in P$ and $\sigma \in N$, $\rho \times \sigma = 0$. So, it is possible to find a way to reset run from ρ with σ. And it is impossible to have lemma as 6.3. Moreover, it was impossible in [BO17], but we have that a run without negative loop is a linear run.

This leads to another interesting question: can we reduce the height of the bridge? To address this question, we rely on Lemma 4.4, which allows us to conclude that it is sufficient to consider runs whose height is polynomial in p_1. More precisely, we only need to study the case where $p_2 \leq (np_1)^{\left(\frac{n^2+3n+2}{2}\right)}$. Remark that Figure 2 shows that no better asymptotic bound can be obtained.

However, as explained previously, this corresponds to studying an open problem. Consequently, no further progress could be made in this direction.

The following part was developed under the assumption that there are no linear bridges in the initial run, since their presence may alter certain conditions, such as the existence of a linear bound that may become polynomial.

4.3 Hill and valley runs

Now, let us consider a run π of the form $\pi_1 \pm p_2\pi_2 \pm p_2 \cdots \pm p_2\pi_k$ where π_i is a run starting and ending bellow $2 \cdot p_1$ and $counter(\pi_i) < p2 - K \cdot p_1$ (respectively starting and ending above $2 \cdot p_1$ and $counter(\pi_i) > 2 \cdot p_1$).

4.3.1 Nonlinear hill and valley runs

For the following, we will work on hill subruns. Valley subruns are studied symmetrically. By [BO17], as with lemma 4.5, we can group non-linear runs in $6Dn^5$ equivalence classes.

Hence, we have the following theorem.

Theorem 4.7. *Given C and states $s, t \in Q_C$. The*

$$\Phi_{NL-HV}(C, s, t) = \{(x, y, \vec{v}) | (s, x) \to^* (t, y) \textit{ in } C \textit{ without non-linear and linear bridge,}$$

$$\textit{where } \vec{v} = \gamma \textit{ and } \gamma(p_2) > n \cdot \gamma(p_1)\}$$

is $\exists$PAD-*definable.*

Proof. As previously, we have the following formula:

$$\begin{aligned} \Phi'_{NL-HV}(C, s, t, l, \vec{s}, \vec{t})(x, y, \vec{p}) = \exists \vec{x}, \vec{y}, & \\ & \bigwedge_{i=1 \ldots l} (2 \cdot p_1 \leq x_i < 3 \cdot p_1 \wedge 2 \cdot p_1 \leq y_{i-1} < 3 \cdot p_1) \\ & \quad \wedge \Phi_0(C_{>p_1, <p_2}, t_{i-1}, s_i)(y_{i-1}, x_i, \vec{p}) \\ & \bigwedge_{i=1 \ldots l} \Phi_{HV}(C, s_i, t_i)(x_i, y_i, \vec{p}) \\ & \wedge p_2 > n \cdot p_1 > n \cdot c_{max} \end{aligned}$$

As $\exists$PAD sets are closed under finite union, Φ_{NL-HV} is $\exists$PAD definable. □

4.3.2 The other classes of runs

Let show that a run of the form $\pi_1 \pm p_2\pi_2 \pm p_2 \cdots \pm p_2\pi_k$ where π_i is a non pumpable or linear run starting and ending bellow $2 \cdot p_1$ and $counter(\pi_i) < p2 - K \cdot p_1$ (respectively starting and ending above $2 \cdot p_1$ and $counter(\pi_i) > 2 \cdot p_1$).

Lemma 4.8. *([BO17]) Given π a linear or non-pumpable run starting and ending between $2 \cdot p_1$ and $3 \cdot p_1$ (respectively $p_2 - p_1$ and p_2), then there exists an equivalent run π' starting and ending between $2 \cdot p_1$ and $3 \cdot p_1$ (respectively $p_2 - p_1$ and p_2) such that counter$(\pi') < K \cdot p_1$ with K being a constant depending only on C.*

By the above lemma, we can build a new POCA denoted by C_{ud}. A POCA C_{ud} is described as follows:

- $Q_{ud} = Q_C \times \{1, 2\}$;
- If $(q, \pm x, q') \in \to_C$ with $x \in Const \cup \{p_1\}$, then

- $((q, i), \pm x, (q_{bis}, i)) \in \rightarrow_{ud}$,
- $((q_{bis}, 1), \leq K \cdot p_1, (q', 1)) \in \rightarrow_{ud}$,
- $((q_{bis}, 2), \geq K \cdot p_1 + 1, (q', 1)) \in \rightarrow_{ud}$,

- If $(q, \pm p_2, q') \in \rightarrow_C$ then $((q, 1), +K \cdot p_1 + 1, (q', 2))$ or $((q, 2), -K \cdot p_1 - 1, (q', 1)) \in \rightarrow_{ud}$.

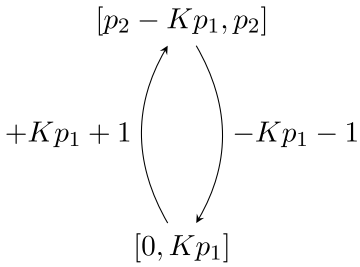


Figure 6: Example of POCA without nonlinear subrun.

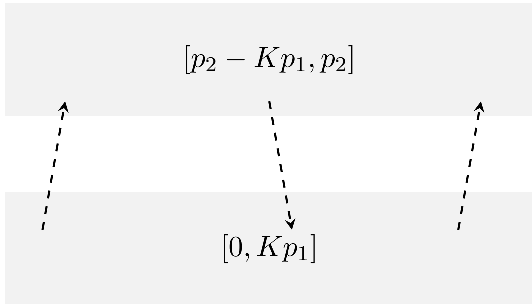


Figure 7: Counter representation

Lemma 4.9. *Given C a POCA without non-linear subruns and bridge subruns, $a, b \in \{0, 1\}$ and C_{ud} the upside down POCA associated to C.*

These two following properties are equivalents.

- *There exists a run π from $(s, a \cdot p_2 + x)$ to $(t, b \cdot p_2 + y)$ in C.*
- *There exists a run π' from $(s, a \cdot (2 \cdot K \cdot p_1 + 1) + x)$ to $(t, b \cdot (2 \cdot K \cdot p_1 + 1) + y)$ in C_{ud}.*

Thus, we obtain a reduction of the parameter p_2 that is linear in p_1.This yields the following theorem.

Theorem 4.10. *Given C and states $s, t \in Q_C$. The*

$$\Phi_{HV}(C, s, t) = \{(x, y, \vec{v}) | (s, x) \rightarrow^* (t, y) \text{ in } C \text{ without non-linear subruns and linear bridges,}$$
$$\text{where } \vec{v} = \gamma \text{ and } \gamma(p_2) > n \cdot \gamma(p_1)\}$$

is $\exists$PAD*-definable.*

Proof. We will use Φ_1 to express Φ_{HV}. This is possible because C contains no $\pm p_2$ transitions and only guards of the form $\bowtie p_1$. □

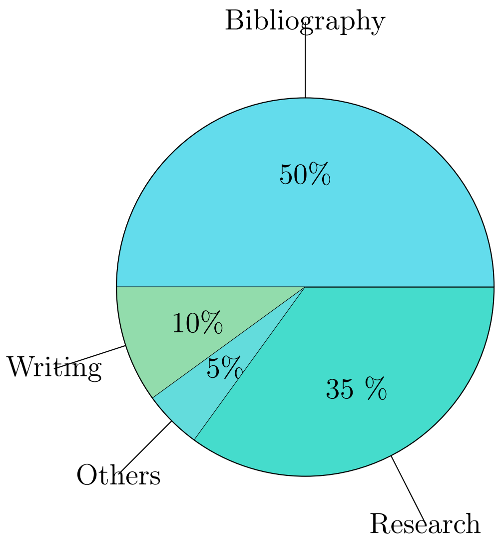
5 Conclusion

After studying POCAs with two parameters, we found it useful to first consider a restricted class. We focus on two-parameter POCAs whose counter is polynomially bounded by the smallest parameter. This restriction provides a better understanding of the general case.

The existing proofs do not completely solve the original problem. However, they clarify its structure and highlight the main difficulties. In particular, neither the reduction to the satisfiability of an $\exists$PAD formula proposed in [BO17], nor the reduction from k to $k-1$ parameters studied in [GH20], is sufficient to establish decidability.

This naturally raises the following open question: is the emptiness problem decidable for a POCA with two parameters p and p^2? To the best of our knowledge, no result in the literature addresses this specific class of POCAs. Understanding this case could be an important step toward solving the general problem. However, the existing techniques do not seem powerful enough to overcome the remaining obstacles.

As requested, we also report the distribution of our working time. The following diagram summarizes the time allocated to each part of the project.



Thank you to Isa for his help and kindness.

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6 Appendix

6.1 Proofs

This section contains in the first part all intermediary proofs. After, There are all gadgets used for the reduction from PTA to POCA.

6.1.1 Remove $+[0, p_i]$ transitions

Theorem 3.1. *Let C be a POCA with* mod$c$ *transitions Then*

$$\Phi_{\bmod c}(C, s, t) = \{(x, y, \vec{q}) | \exists \pi : (s, x) \to (t, y) \in C \text{ where } \vec{p} = \vec{q}\}$$

is $\exists PAD$ definable.

Proof. Given $D \in [\mathbb{Z}_{c_1} \times \cdots \times \mathbb{Z}_{c_r}]^k$, we construct the following formula $M'(A, s, t, D)(x, y, \vec{p})$ asserting that D is consistent with $\vec{p}$ and that $(s, x) \to (t, y)$ in C_{mod}:

$$\begin{aligned}\Phi'_{\bmod c}(A, s, t, D)(x, y, \vec{p}) = \exists \overrightarrow{v(r)}, \overrightarrow{w(r)}, \\ \bigwedge_{i=1\ldots|Const_C|} x \equiv v(i) \bmod c_i \wedge y \equiv w(i) \bmod c_i \wedge p(j) \equiv D(j)(i) \bmod c_i \\ \wedge\, \Phi_{+[0,p_i]}(C_{mod}, s \times \overrightarrow{v(r)}, t \times \overrightarrow{w(r)})(x, y, \vec{p})\end{aligned}$$

Where $\Phi_{+[0,p_i]}$ is a formula from 3.4. Moreover, the expression $a \equiv b \bmod c$ can be encoded in ∃PAD by $a - b | c$.
Since ∃PAD definability is closed under finite union and there are only finitely many D, we can conclude that $\Phi_{\bmod c}(C, s, t)$ is ∃PAD definable. □

Lemma 3.2. *Let C be a parametric bounded one-counter machine. If there is an accepting bounded run in C, then there is an accepting bounded run π such that at most $2 \cdot n \cdot R$ counter increments by $+[0, p_i]$ transitions are not in the set $\{+0, +1, \ldots, +R, +p_1 - R, \ldots, +p_1, \ldots, +p_r - R, \ldots, +p_r\}$. That is,*

$$|\{\, i \mid \exists j \leq r\ , \pi(i) = +[0, p_j] \text{ and } R < \text{counter}(\pi(i+1)) - \text{counter}(\pi(i)) < p_j - R\}| \leq 2 \cdot n \cdot R \cdot r.$$

For every $+[0, p_i]$, we will show that all but a bounded number of $+[0, p_i]$ transitions increment the counter by at most R or by at least $p_i - R$. Suppose that there are at least two $+[0, p_i]$ transitions along π. Then we can write π as $\tau_1 \xrightarrow{e_1} \tau_2 \xrightarrow{e_2} \tau_3$ where e_1 and e_2 are $+[0, p_i]$ transitions. Denote the counter updates at e_1 and e_2 by f_1 and f_2, respectively. If $R < f_1, f_2 < p_i - R$, then consider the run τ_2' which is obtained from τ_2 by incrementing the counter by R. That is, for every index i, we have $\text{state}(\tau_2(i)) = \text{state}(\tau_2'(i))$ and $\text{counter}(\tau_2'(i)) = \text{counter}(\tau_2(i)) + R$. This corresponds to incrementing the counter in e_1, e_2 by $f_1 + R, f_2 - R$ respectively. By [BO17], this transformation is possible. Repeating the same argument yields the desired result.

We use the above observations to repeatedly increment some transition $e_1 = +[0, p]$ and simultaneously decrement some other transition $e_2 = +[0, p]$ by R, eventually stopping when no such pair of transitions exists. Formally, let e_1 be the first $+[0, p]$ transition such that there exists a transition $e_2 = +[0, p]$. Then e_1 can be incremented by R and e_2 can be decremented by R. Let the run π be of the form $\pi = \tau_1 \xrightarrow{e_1} \tau_2 \xrightarrow{e_2} \tau_3$. Replace π by the modified run π' and repeat the process, thereby creating the sequence of runs π, π', π'', π''', … Note that $\text{counter}(\pi')$ is lexicographically larger than $\text{counter}(\pi)$. Since π has finite length and every transition can be incremented by at most $|\pi|\, p$, the procedure eventually terminates after some run $\pi^{(k)}$. Let $\{f_1, \ldots, f_l\}$ be the set of increments caused by $+[0, p]$-transitions along $\pi^{(k)}$. By the above argument, each f_i is either at most R, at least $p - R$, or one of at most nR distinct intermediate values.

Moreover, we can repeat this process for each p_i such that there exists $+[0,p_i]$ transition. Then $\pi^{(k)}$ can be factored as $\pi^{(k)} = \pi_0 \to \pi_1 \to \cdots \to \pi_v$ where each π_i is a run in a POCA obtain from C where each $+[0,p_i]$ transition is replaced by $2 \cdot R + 2$ transitions: $+0, +1, \ldots, +R, +p_i - R, \ldots, +p_i$ and there is a $+[0,p_j]$ transition between π_i and π_{i+1}. Remark that $l \leq 2 \cdot n \cdot R \cdot |P|$.

Theorem 3.4. *Given a POCA C with $\equiv 0 \pmod{c}$ and $+[0,p_i]$ transitions, let $s, t \in C$ be states. Then the set $\Phi_{+[0,P]}(C,s,t) = \{(x,y,\vec{q}) \mid \exists \pi : (s,x) \to^* (t,y) \text{ in } C,\ \vec{p} = \vec{q}\}$ is $\exists$PAD-definable if, and only if, reachability for C without* mod$c$ *and $+[0,p_i]$ is $\exists$PAD-definable.*

Proof. Let G be a POCA obtained from C by replacing each $+[0,p_i]$ transition by $2 \cdot R + 2$ transitions: $+0, +1, \ldots, +R, +p_i - R, \ldots, +p_i$. Then as describe above, $\pi^{(k)} = \pi_0 \to \pi_1 \to \cdots \to \pi_v$. Let $v \leq 2 \cdot n \cdot R \cdot |P|$ and states $\vec{s} = (s_1, \ldots, s_l)$ and $\vec{t} = (t_1, \ldots, t_l)$ and parameters $\vec{q} = (q_1, \ldots, q_{l-1})$.

$$\begin{aligned}
\Phi'_{+[0,P]}(C,s,t,v,\vec{s},\vec{t},\vec{q})(x,y,\vec{p}) = \exists \vec{x}, \vec{y},& \\
\bigwedge_{i=1\ldots l} \Phi_{NL-B}(C,s_i,t_i)(x_i,y_i,\vec{p})& \\
\bigwedge_{i=1\ldots l-1} 0 \leq y_{i+1} - x_i \leq p_i \wedge \Phi_1(t_i, +[0,p_i], s_{i+1})&
\end{aligned}$$

Since $l \leq 2 \cdot n \cdot R \cdot |P|$ there are finitely many possibilities for $\vec{s}$, $\vec{s}$ and $\vec{q}$. As $\exists$PAD sets are closed under finite union, $\Phi_{+[0,P]}$ is $\exists$PAD definable. □

6.1.2 A better bound

Lemma 3.5. *Given C a POCA and C_b is POCA as define above. These two following propositions are equivalent:*

- *There exists a run π from (s,x) to (t,y) in C.*
- *There exists a run π' from $(s, x \mod p_2)$ to $(t, y \mod p_2)$ in C_b.*

Proof. Let C be a POCA and let C_b be the POCA constructed from C, whose counter is bounded by $p_1 + p_2$.

($\Rightarrow$) Assume that there exists a run π in C from (s,x) to (t,y).

Define a mapping f on configurations of C as follows: $f(s, a \cdot p_2 + b) = \begin{cases} (s_{<p_2}, b) & \text{if } a = 0, \\ (s_{\geq p_2}, b) & \text{otherwise,} \end{cases}$ where $0 \leq b < p_2$.

By construction of C_b, every transition of π is simulated by a corresponding transition in C_b. Hence, applying f component wise to the run π yields a valid run $f(\pi)$ in C_b from $(s, x \bmod p_2)$ to $(t, y \bmod p_2)$.

($\Leftarrow$) Conversely, assume that there exists a run π' in C_b from $(s, x \bmod p_2)$ to $(t, y \bmod p_2)$.

We reconstruct a run π in C by lifting each configuration of π' to a concrete counter value consistent with its residue modulo p_2. More precisely, each configuration (s', z) of C_b is interpreted as some configuration $(s', a \cdot p_2 + z)$ in C, where a is chosen so that the counter updates of C_b correspond to valid updates in C.

By construction of C_b, every transition of C_b corresponds to a transition of C that preserves the counter value modulo p_2 and is compatible with the bounded encoding of the state. Hence, each step of π' can be simulated in C without violating the semantics of the original machine.

Therefore, we obtain a valid run π in C from (s,x) to (t,y).

□

6.1.3 Linearly bounded by all parameters

Lemma 6.1. *Let s_1, s_2 be states of C, and let $0 \le q_1, q_2 < k_n$. Assume moreover that $M < r_1, r_2 < p_n - M$. Then the following are equivalent:*

- *There exists a run π in C from $\Big(s_1, q_1 p_n + r_1\Big)$ to $\Big(s_2, q_2 p_n + r_2\Big)$ such that $\Delta(\pi) \le k_n p_n$ and $\Delta(\pi) \bmod p_n \in (M, p_n - M)$.*
- *There exists a run π' in C_{mod} from $((s_1, q_1), r_1)$ to $((s_2, q_2), r_2)$.*

Proof. ($\Rightarrow$) Assume that such a run π exists. Since $\Delta(\pi) \bmod p \in (M, p_n - M)$, every configuration along π admits a unique decomposition of the form $\Delta = qp_n + r$ with $0 \le q < k_n$ and $M < r < p_n - M$.

Define a mapping f that sends a configuration $\Big(s, qp_n + r\Big)$ of C to$((s, q), r)$ in C_{mod}. By construction of C_{mod} and since the run never leaves the central interval $(M, p_n - M)$, the image $f(\pi)$ is a valid run in C_{mod} from $((s_1, q_1), r_1)$ to $((s_2, q_2), r_2)$.

($\Leftarrow$) Conversely, assume that a run π' exists in C_{mod}. Since $M < r_1, r_2 < p_n - M$, by Lemma 8.3 in [BO17] we may assume that $\mathsf{counter}(\pi') \subseteq (M, p_n - M)$. Define a mapping g that sends a configuration $((s, q), r)$ of C_{mod} to $\Big(s, qp_n + r\Big)$ in C. By construction of C_{mod}, the image $g(\pi')$ is a valid run in C from $\Big(s_1, q_1 p_n + r_1\Big)$ to $\Big(s_2, q_2 p_n + r_2\Big)$. Moreover, along this run we have $\Delta(g(\pi')) \le k_n p_n$ and $\Delta(g(\pi')) \bmod p_n \in (M, p_n - M)$. □

Lemma 6.2. *Given $k_n \in \mathbb{N}$, states s_1, s_2 of C, numbers $0 \le q_1, q_2 < k_n$ and $r_1, r_2 \in [0, M] \cup [p_n - M, p_n)$, one can decide in polynomial time whether there exists a run π from $(s_1, q_1 p_n + r_1)$ to $(s_2, q_2 p_n + r_2)$ in C such that $\Delta(\pi) \bmod p \in [0, M] \cup [p_n - M, p_n)$ for every parameter p.*

Proof. As in Lemma 8.5 in [BO17], the existence of such runs π is independent of the actual values p_n and hence independent of all parameters; it can therefore be precomputed. □

Theorem 4.3. *Given C with n parameters, $k_i \in \mathbb{N}$, and states s, t of C, the set*

$$\Phi_2(C, s, t, k_1, \ldots, k_n) = \{(x, y) \mid \exists \pi : (s, x) \to (t, y) \text{ and } \Delta(\pi) \le k_i p_i \text{ for all } i\}$$

is $\exists$PAD-definable.

Proof. Let c_1, c_2 be configurations of C and let $\pi : c_1 \to c_2$ be a shortest run. Then π can be decomposed into subruns close to and far from multiples of all parameters of C, that is, $\pi = \tau_0 \to \pi_1 \to \tau_1 \to \cdots \to \pi_\ell \to \tau_\ell$ such that for every τ_i we have $\Delta(\tau_i) \bmod p_n \in [0, M] \cup [p_n - M, p_n)$ and we have $\Delta(\pi_i) \bmod p_n \in (M, p_n - M)$. The configuration $first(\tau_i)$ is uniquely determined by the state of C, the value q_n with $0 \le q_n < k_n$, and values $\Delta(first(\tau_i)) \bmod p_n$, which admits at most $2M + 1$ chances; hence the number of possible initial configurations of τ_i is bounded by $|S|k_n \cdot (2M + 1)$ and thus $\ell = \mathcal{O}(|S|k_n \cdot M)$. By Lemma 6.1, the existence of each π_i is witnessed in C_{mod}, and by Lemma 6.2 the existence of each τ_i is independent of p_n and can be precomputed. Reachability in C_{mod} is $\exists$PAD-expressible by the induction hypothesis. Taking the conjunction of the corresponding $\exists$PAD formulas yields a single $\exists$PAD formula

$$\Psi_n \wedge \bigwedge_{i=0}^{\ell} \Phi(C_{\text{mod}}, first(\pi_i), last(\pi_i))(\Delta(first(\pi_i)), \Delta(last(\pi_i)), p_n),$$

where Ψ_n encodes the validity of the τ_i segments and the consistency between consecutive τ_i and π_i. By construction, the counter of C_{mod} is bounded by $k_{n-1} p_{n-1}$. Therefore, by the induction hypothesis, there exists an $\exists$PAD formula $\Phi(C_{\text{mod}}, s, t)$ defining the reachability relation of C_{mod}, since C_{mod} has the $n - 1$ parameters $p_1, \ldots, p_n$. Since there are only finitely many possible initial and final configurations of the τ_i and π_i, and these uniquely determine the decomposition, taking the disjunction over all such factorization yields an $\exists$PAD formula defining the reachability relation. □

6.1.4 Non-linear bridges

Lemma 6.3. *Let π be a nonlinear run between (s, x) and (t, y) such that $2 \cdot p_1 < x \leq 3 \cdot p_1$ and $p_2 - 2 \cdot p_1 \leq y < p_2 - p_1$. Then, for every $k \in \mathbb{Z}$ with $y + k \cdot L \geq p_2 - 2 \cdot p1$, there is a run τ between (s, x) and $(t, y + k \cdot L)$ with $L \in \mathbb{N}$.*

Proof. Write $\pi = \pi_1 \cdot \pi_2 \cdot \pi_3 = \pi_1 \rightarrow \alpha a \rightarrow f+$ a negative loop σ such that $\rho \times \sigma \neq 0$. Write $\Delta(\rho) = a \cdot p_1 + b$ and $\Delta(\sigma) = c \cdot p_1 + d$. Then $c \leq 0 \leq a$, and $a \cdot \Delta(\sigma) - c \cdot \Delta(\rho) = a(cp_1 + d) - c(ap_1 + b) = ad - cb = \rho \times \sigma$. Hence $a \cdot k \cdot \Delta(\sigma) - c \cdot k \cdot \Delta(\rho) = k(\rho \times \sigma)$ for any $k \in \mathbb{N}$. Denote $\rho \times \sigma$ by P, and suppose that the path π takes the loops ρ and σ exactly r and s times, respectively.

First assume that $P > 0$. Then for any $k \in \mathbb{N}$, by Lemma 4.4 there exists a path π_k obtained by modifying the traversal counts as $\pi_k = \pi(s \leftarrow s + ak,\ r \leftarrow r + |c|k)$ such that $\text{counter}(\pi_k) > 2p_1$ and $\Delta(\pi_k) = \Delta(\pi) + kP$. Thus $\text{counter}(\text{last}(\pi_k)) = y + kP$, and changing the number of traversals of ρ and σ allows us to increase the counter by arbitrary multiples of P.

Recall that $\Delta(\sigma) < 0$. Therefore there exists a largest $q \in \mathbb{N}$ such that $qP + \Delta(\sigma) < 0$, hence $-P \leq qP + \Delta(\sigma) < 0$. Let $R = qP + \Delta(\sigma)$. Then

$$\begin{aligned} qa \cdot \Delta(\sigma) - c \cdot \Delta(\rho) + \Delta(\sigma) &= q(\rho \times \sigma) + \Delta(\sigma) \\ &= qP + \Delta(\sigma) \\ &= R. \end{aligned}$$

For $k \in \mathbb{N}$, suppose that $y + kR \geq p_2 - 2p_1$. Since $p_2 - p_1 > y$, it follows that $kR > -p_1$. Hence, by Lemma 4.4, there exists a path $\tau_k = \pi(s \leftarrow s + k(qa+1),\ r \leftarrow r + |c|qk)$ such that $\Delta(\tau_k) = \Delta(\pi) + kR$ and $\text{counter}(\tau_k) + kR > 2p_1 + kR > p_1$.

Therefore $\text{counter}(\text{last}(\tau_k)) = y + kR$.

Finally, let $L = \text{lcm}(P, |R|)$ be the least common multiple of P and $|R|$. Then $L \leq PR \leq P^2 \leq n^4$, and for every $k \in \mathbb{Z}$ such that $y + kL > p_2 - 2 \cdot p_1$, there exists a valid run from (s, x) to $(t, y + kL)$ in C^γ.

The case $P = \rho \times \sigma < 0$ is symmetric. □

Now, there are $O(2^n)$ equivalence classes. So consider nonlinear run $\pi_1, \pi_2, \ldots$ ending starting in $(s_1, x_1), (s_2, x_2), \ldots$ and ending in $(t_1, y_1), (t_2, y_2), \ldots$ where $2 \cdot p_1 < x_i \leq 3 \cdot p_1$ and $p_2 - 2 \cdot p_1 \leq y_i < p_2 - p_1$. We have that $p_i \sim_L p_j$ if, and only if, we have $y_i \equiv y_j \mod L$. Then, using the above lemma, there is a run from (s_i, x_i) to (t_j, y_j). And so, we can skip all π_k for $i < k < j$.

Lemma 4.5. *Given a run $\pi = \sigma_0 \pi_1 \sigma_1 \pi_2 \ldots \pi_k \sigma_k$ where σ_i is a run starting and ending bellow $2 \cdot p_1$ (above $p_2 - p_1$) when the run stays bellow $p_2 - p_1$ (or above $2 \cdot p_1$) and π_i is a run starting bellow $2 \cdot p_1$ and ending above $p_2 - p_1$ (or starting above $2 \cdot p_1$ and ending bellow $p_2 - p_1$). Then, there exists an equivalent run $\pi' \sim \pi$ such that π' factors as:*

$$\sigma'_0 \pi'_1 \sigma'_1 \pi'_2 \ldots \pi'_m \sigma'_{g(m)}$$

where

- $\pi'_i : first(\pi_{f(i-1)}) \rightarrow last(\pi_{f(i)})$ *if* π'_i *is a nonlinear run;*
- $g : [1 \ldots l] \rightarrow [1 \ldots k]$ *is a strictly increasing function;*
- $m \leq 2 \cdot D \cdot n^5$.

Proof. The following proof follows [BO17] closely, except that the downward bridges must also be considered, in addition to the upward ones. □

Theorem 4.6. *Given C and states $s, t \in Q_C$. The*

$$\Phi_{NL-B}(C, s, t) = \{(x, y, \vec{v}) | (s, x) \rightarrow^* (t, y) \text{ in } C \text{ without linear bridge,}$$

$$\text{where } \vec{v} = \gamma \text{ and } \gamma(p_2) > n \cdot \gamma(p_1)\}$$

is ∃PAD-definable.

Proof. We encode the existence accepting runs from lemma 4.5. Given $l \in \mathbb{N}$ the length of the factoring in lemma 4.5 and states $\vec{s}, \vec{t} \in Q_C^{m+1}$ the initial and final states of π'_i, respectively, consider the following formula:

$$\Phi'_{NL-B}(C, s, t, m, \vec{s}, \vec{t})(x, y, \vec{p}) = \exists \vec{x}, \vec{y},$$

$$\bigwedge_{i=1\ldots m} \Big((2 \cdot p_1 \leq x_i < 3 \cdot p_1 \wedge p_2 - 2 \cdot p_1 < y_{i-1} \leq p_2 - p_1)$$

$$\vee\, (p_2 - 2 \cdot p_1 < x_i \leq p_2 - p_1 \wedge 2 \cdot p_1 \leq y_{i-1} < 3 \cdot p_1)\Big)$$

$$\wedge\, \Phi_0(C_{>p_1,<p_2}, t_{i-1}, s_i)(y_{i-1}, x_i, \vec{p})$$

$$\bigwedge_{i=1\ldots m} \Phi_{NL-HV}(C, s_i, t_i)(x_i, y_i, \vec{p})$$

$$\wedge\, p_2 > n \cdot p_1 > n \cdot c_{max}$$

Where we denote by $C_{>p_1,<p_2}$, the one-counter automata obtained from C by removing all edges $\leq p_1$ and $\geq p_2$ and testing that the counter is between $p1$ and p_2 after every update. As ∃PAD sets are closed under finite union, $\Phi_{+[0,P]}$ is ∃PAD definable. □

6.1.5 The other classes of runs

Lemma 4.9. *Given C a POCA without non-linear subruns and bridge subruns, $a, b \in \{0, 1\}$ and C_{ud} the upside down POCA associated to C.*

These two following properties are equivalents.

- *There exists a run π from $(s, a \cdot p_2 + x)$ to $(t, b \cdot p_2 + y)$ in C.*
- *There exists a run π' from $(s, a \cdot (2 \cdot K \cdot p_1 + 1) + x)$ to $(t, b \cdot (2 \cdot K \cdot p_1 + 1) + y)$ in C_{ud}.*

Proof. Given $a, b \in \{0, 1\}$, C and C_{ud}.

- $(\Rightarrow)$. Let π a run in C from $(s, a \cdot p_2 + x)$ to $(t, b \cdot p_2 + y)$. Then, we can factorize π as $\pi_1 \cdot \pm p_2 \cdot \pi_2 \cdot \pm p_2 \cdot \pi_3 \cdots$. By lemma 4.8, for every π_i, there exists a run π'_i bounded by $K \cdot p_1$ for hill-run and $p_2 - K \cdot p_1$ for valley-run.

  Hence, all run between $[0, K \cdot p_1]$ are accepting run in $Q_C \times \{0\}$ and all run between $[p_2 - K \cdot p_1, p_2 + p_1]$ are accepting run in $Q_C \times \{1\}$. And, as in construction, every $\pm p_2$ transitions are replaced by $2 \times K$ transitions $\pm p_1$ and one ± 1 transition. We conclude that there exists a run $(s, a \cdot (2 \cdot K \cdot p_1 + 1) + x)$ to $(t, b \cdot (2 \cdot K \cdot p_1 + 1) + y)$ in C_{ud}.
- $(\Leftarrow)$ Let π' a run from $(s, a \cdot (2 \cdot K \cdot p_1 + 1) + x)$ to $(t, b \cdot (2 \cdot K \cdot p_1 + 1) + y)$ in C_{ud}. Then, we can factorize π as $\pi_1 \cdot \pi_2 \cdot \pi_3 \cdots \pi_k$ where π_{2i} are in $Q_C \times \{1 - a\}$ and π_{2i+1} are in $Q_C \times \{a\}$. By construction of C_{ud}, every run in $Q_C \times \{1 - a\}$ with $a = 1$ (respectively $a = 0$) has its counter belongs $[0, K \cdot p_1]$ (respectively $[K \cdot p_1 + 1, 2 \cdot K \cdot p_1 + 1]$).

  Hence, for every run π_{2i} if $a = 0$, then there exists an equivalent run in C between $[p_2 - K p_1, p_2]$ else in $[0, K p_1]$. It is the same think for every run π_{2i+1}. Then, we conclude that there exists a run π from $(s, a \cdot p_2 + x)$ to $(t, b \cdot p_2 + y)$ in C.

□

6.2 PTA to POCA reduction

In this subsection, we provide a detailed description of the gadgets used in the reduction from PTAs to POCAs, as outlined in Theorem 1.4. The construction from [BO17] uses the difference between the two clocks to encode the value of the counter. And the value of the counter is updated when one of the two clocks is reset. An update depends on the region in which the system is.

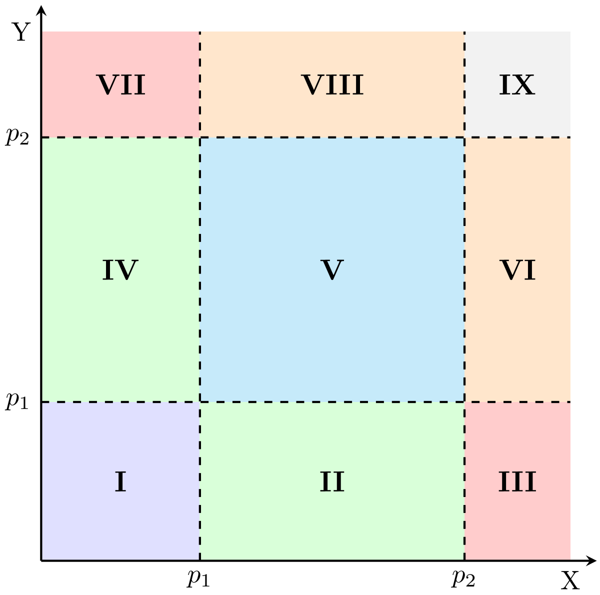

Figure 8: Schema of regions

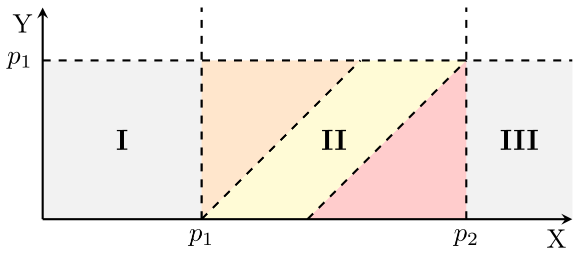

Figure 9: Schema of region II

A region is defined by the set of clock valuations between two parameters. For example, region II from 8 is the set of clock valuations $[p_1, p_2] \times [0, p_1]$. The gadgets used to update the counter depend on the counter's location and on which clock must be reset. For example, if the counter is in region II in the yellow part (middle) of 9 and that we want to reset the y clock, then we will use the gadget from 16.

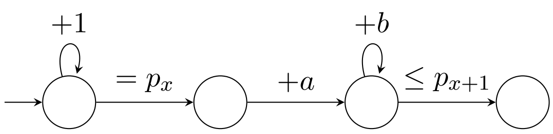

Figure 10: Reset y from left to left

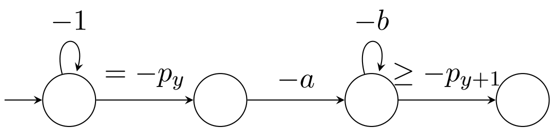

Figure 11: Reset x from bottom to bottom

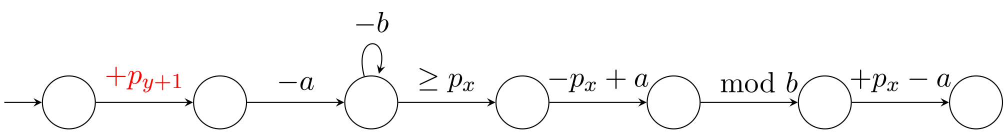

Figure 12: Reset y from left to bottom

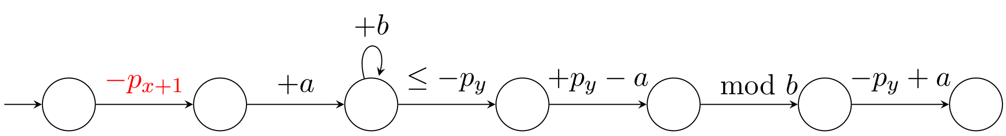

Figure 13: Reset x from bottom to left

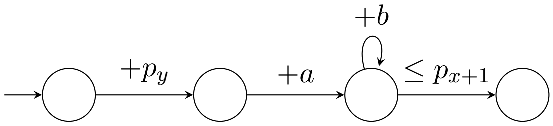

Figure 14: Reset y from bottom to left

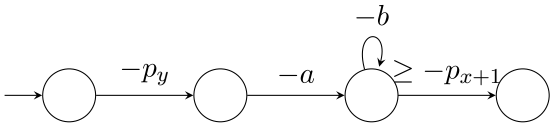

Figure 15: Reset x from left to bottom

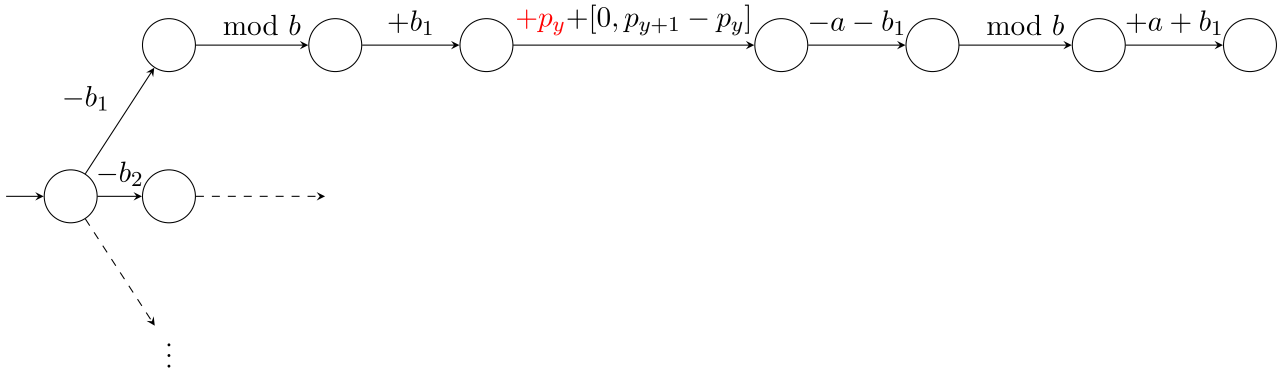

Figure 16: Reset y from bottom to bottom

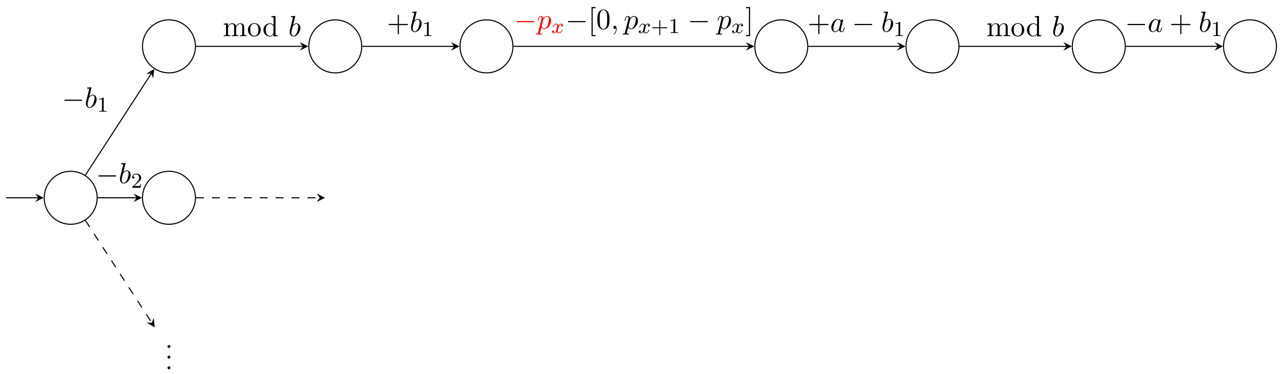

Figure 17: Reset x from left to left